

7.4.40

EE25BTECH11033 - Kevin

Question:

Let a circle be given by $2x(x - a) + y(2y - b) = 0$, ($a \neq 0, b \neq 0$). Find the condition on a and b if two chords, each bisected by the X axis, can be drawn to the circle from $(a, \frac{b}{2})$.

Solution:

The equation of the circle is ,

$$2x(x - a) + y(2y - b) = 0 \quad (1)$$

$$\implies 2x^2 - 2ax + 2y^2 - by = 0 \quad (2)$$

$$\implies x^2 + y^2 - ax - \frac{b}{2}y = 0 \quad (3)$$

The chords are drawn from the point $\mathbf{P}(a, \frac{b}{2})$, which lies on the circle since

$$a^2 + \left(\frac{b}{2}\right)^2 - a(a) - \frac{b}{2}\left(\frac{b}{2}\right) = a^2 + \frac{b^2}{4} - a^2 - \frac{b^2}{4} = 0 \quad (4)$$

Let a chord start at $\mathbf{P}(a, \frac{b}{2})$ and end at another point on the circle, $\mathbf{Q}(x_q, y_q)$. The midpoint of this chord is $\mathbf{M} = \left(\frac{a+x_q}{2}, \frac{b/2+y_q}{2}\right)$.

Since this chord is bisected by the X -axis,

$$\frac{b/2 + y_q}{2} = 0 \implies \frac{b}{2} + y_q = 0 \implies y_q = -\frac{b}{2} \quad (5)$$

Since $\mathbf{Q}(x_q, -\frac{b}{2})$ must also lie on the circle,

$$x_q^2 + \left(-\frac{b}{2}\right)^2 - a(x_q) - \frac{b}{2}\left(-\frac{b}{2}\right) = 0 \quad (6)$$

$$x_q^2 + \frac{b^2}{4} - ax_q + \frac{b^2}{4} = 0 \quad (7)$$

$$x_q^2 - ax_q + \frac{b^2}{2} = 0 \quad (8)$$

The problem requires that two such chords can be drawn. This implies that the quadratic equation for x_q must have two distinct real roots.

Any monic quadratic equation of the form $\lambda^2 - S\lambda + P = 0$ can be interpreted as the **characteristic equation** of a 2×2 matrix **A**, where S is the trace and P is the determinant of the matrix.

$$|\mathbf{A} - \lambda \mathbf{I}| = \lambda^2 - \text{tr}(\mathbf{A})\lambda + |\mathbf{A}| = 0 \quad (9)$$

The roots of this equation are the **eigenvalues** of the matrix **A**.

Here,

- $\lambda = x_q$
- $\text{tr}(\mathbf{A}) = a$
- $|\mathbf{A}| = \frac{b^2}{2}$

A fundamental property of a 2×2 matrix is that it has **two distinct real eigenvalues** if and only if the discriminant of its characteristic equation is positive. This condition, in terms of its trace and determinant, is:

$$(\text{tr}(\mathbf{A}))^2 - 4|\mathbf{A}| > 0 \quad (10)$$

With $\text{tr}(\mathbf{A}) = a$ and $|\mathbf{A}| = \frac{b^2}{2}$, the condition becomes:

$$a^2 - 4\left(\frac{b^2}{2}\right) > 0 \quad (11)$$

$$a^2 - 2b^2 > 0 \quad (12)$$

$$a^2 > 2b^2 \quad (13)$$

This is the required condition on a and b .

For $a = 2$ and $b = 1$,

